Linux Basic Commands –

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1. File & Directory Commands

Command	Full Form	Meaning	Example
`pwd`	Print Working Directory	Shows where you are currently	`pwd`
`ls`	List	Shows files inside a folder	`ls` , `ls -l` , `ls -a`
`cd`	Change Directory	Move to another folder	`cd /home` , `cd ..`
`mkdir`	Make Directory	Create a new folder	`mkdir newfolder`
`rmdir`	Remove Directory	Delete an empty folder	`rmdir oldfolder`
`rm`	Remove	Delete file or folder	`rm file.txt`
`cp`	Copy	Copy file or folder	`cp file1.txt file2.txt`
`mv`	Move	Move or rename file/folder	`mv old.txt new.txt`
`touch`	-	Create an empty file	`touch newfile.txt`
`find`	Find	Search for files	`find /home -name "*.txt"`

Examples Explained:

```
# Show current location
pwd
# Output: /home/user

# List files
```

```
ls
ls -l       # Detailed list
ls -a       # Show hidden files

# Change directory
cd /home        # Go to /home folder
cd ..           # Go one folder back
cd ~            # Go to home folder

# Create folder
mkdir newfolder

# Delete empty folder
rmdir oldfolder

# Delete file
rm file.txt
rm -rf foldername    # Force delete folder with all files inside

# Copy file
cp file1.txt file2.txt
cp -r folder1 folder2   # Copy entire folder

# Move or Rename
mv old.txt new.txt       # Rename file
mv file.txt /home/       # Move file to /home

# Create empty file
touch newfile.txt

# Find files
find /home -name "*.txt"  # Find all .txt files in /home
```

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2. File Viewing & Editing Commands

Command	Meaning	Example
`cat`	Show file content	`cat file.txt`

less	View file by scrolling	less bigfile.log
head	Show first 10 lines	head file.txt
tail	Show last 10 lines	tail file.txt
nano	Simple text editor	nano file.txt
vi/vim	Advanced text editor	vi file.txt

Examples:

```
# Display file content
cat file.txt

# Scroll through large file (press q to exit)
less bigfile.log

# Show first 10 lines
head file.txt

# Show last 10 lines
tail file.txt

# Watch live log updates
tail -f logfile.log

# Open file in simple editor
nano file.txt

# Open file in advanced editor
vi file.txt
```

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3. Permission & Ownership Commands

Concept:

```
Linux has 3 types of permissions:
   Read   (r) = Can view file
   Write  (w) = Can edit file
   Execute(x) = Can run file

3 types of users:
   User  (owner)
   Group
   Others (everyone else)
```

Command	Meaning	Example
`chmod`	Change file permissions	`chmod 755 script.sh`
`chown`	Change file owner	`chown user:group file.txt`
`sudo`	Run command as admin	`sudo apt update`

Permission Numbers – Easy to Remember:

```
7 = Read + Write + Execute  (rwx)     Full access
6 = Read + Write            (rw-)     Can read & edit
5 = Read + Execute          (r-x)     Can read & run
4 = Read Only               (r--)     Can only read
0 = No permission           (---)     No access
```

Examples:

```
# Give full permission to owner, read+execute to others
chmod 755 script.sh

# Make file executable
chmod +x file.sh

# Remove write permission
chmod -w file.txt

# Change owner of file
```

```
chown john file.txt
chown john:developers file.txt

# Run command with admin power
sudo apt update
```

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4. User Management Commands

Command	Meaning	Example
`whoami`	Shows current username	`whoami`
`id`	Shows user ID and group ID	`id username`
`adduser`	Create a new user	`sudo adduser john`
`passwd`	Change password	`passwd`
`usermod`	Modify user settings	`sudo usermod -aG sudo john`
`su`	Switch to another user	`su - root`
`who`	Show who is logged in	`who`

Examples:

```
# Who is current user?
whoami
# Output: john

# Show user info
id john
# Output: uid=1001(john) gid=1001(john)

# Create new user
sudo adduser john

# Change your own password
passwd
```

```
# Change another user's password
sudo passwd john

# Add user to sudo group
sudo usermod -aG sudo john

# Switch to root user
su - root

# See who is logged in
who
```

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5. System Information Commands

Command	Meaning	Example
`uname -a`	Show OS and kernel info	`uname -a`
`df -h`	Show disk space	`df -h`
`du -sh`	Show folder size	`du -sh /home`
`free -h`	Show RAM usage	`free -h`
`top/htop`	Live CPU & RAM usage	`top`
`uptime`	How long system is running	`uptime`
`history`	Show old commands	`history`

Examples:

```
# Show full system info
uname -a
# Output: Linux ubuntu 5.15.0 #1 SMP x86_64 GNU/Linux

# Show disk space (human readable)
df -h
```

```
# Output:
# /dev/sda1   50G   20G   28G   42%  /

# Show folder size
du -sh /home
# Output: 2.5G /home

# Show RAM usage
free -h
# Output:
# Mem:  8.0G   4.5G   3.5G
# Swap: 2.0G   0B     2.0G

# Live process monitor (press q to exit)
top

# How long system has been running
uptime
# Output: 10:00:00 up 5 days, 3:22

# Show last commands used
history
```

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6. Package Management (Install Software)

Task	Ubuntu/Debian	CentOS/RHEL
Update list	`sudo apt update`	`sudo yum update`
Install software	`sudo apt install nginx`	`sudo yum install nginx`
Remove software	`sudo apt remove nginx`	`sudo yum remove nginx`
Search software	`apt search nginx`	`yum search nginx`

Examples:

```
# ===== Ubuntu / Debian =====

# Update package list
sudo apt update

# Install a package
sudo apt install nginx

# Remove a package
sudo apt remove nginx

# Search for a package
apt search nginx

# Update + Install together
sudo apt update && sudo apt install nginx

# ===== CentOS / RHEL =====

sudo yum update
sudo yum install nginx
sudo yum remove nginx
yum search nginx
```

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7. Network Commands

Command	Meaning	Example
`ip addr`	Show IP address	`ip a`
`ping`	Check network connection	`ping google.com`
`ssh`	Connect to remote server	`ssh user@192.168.1.10`
`scp`	Copy file to remote server	`scp file.txt user@ip:/path`
`wget/curl`	Download file from internet	`wget url`
`netstat`	Show open ports	`netstat -tulnp`

`hostname`	Show system name	`hostname`

Examples:

```
# Show IP address
ip a
ifconfig

# Check if internet or server is working
ping google.com
ping 192.168.1.1

# Connect to remote server
ssh user@192.168.1.10
ssh john@server.example.com

# Copy file to remote server
scp file.txt john@192.168.1.10:/home/john/

# Download file from internet
wget https://example.com/file.zip
curl -O https://example.com/file.zip

# Show open ports
netstat -tulnp

# Show computer name
hostname
```

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8. Process Management

Command	Meaning	Example
`ps aux`	Show all running processes	`ps aux`
`kill`	Stop a process by ID	`kill 1234`

`kill -9`	Force stop a process	`kill -9 1234`
`systemctl`	Start/stop services	`systemctl start nginx`
`jobs`	Show background jobs	`jobs`

Examples:

```
# Show all running processes
ps aux

# Find specific process
ps aux | grep nginx

# Stop process (by PID number)
kill 1234

# Force stop process
kill -9 1234

# Run program in background
firefox &

# Service management
systemctl start nginx      # Start nginx
systemctl stop nginx       # Stop nginx
systemctl restart nginx    # Restart nginx
systemctl enable nginx     # Auto-start on boot
systemctl status nginx     # Check status

# Show background jobs
jobs
```

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9. Compression & Archive Commands

Command	Meaning	Example

`tar`	Pack/unpack files	`tar -czvf backup.tar.gz folder/`
`gzip`	Compress to .gz	`gzip file.txt`
`gunzip`	Decompress .gz file	`gunzip file.txt.gz`
`zip`	Compress to .zip	`zip -r file.zip folder/`
`unzip`	Extract .zip file	`unzip file.zip`

Examples:

```
# Create compressed archive
tar -czvf backup.tar.gz folder/

# Extract compressed archive
tar -xzvf backup.tar.gz

# View contents without extracting
tar -tvf backup.tar.gz

# Compress single file
gzip file.txt
# Result: file.txt.gz

# Decompress gz file
gunzip file.txt.gz

# Create zip archive
zip -r myarchive.zip folder/

# Extract zip file
unzip myarchive.zip

# Extract to specific location
unzip myarchive.zip -d /home/user/
```

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10. Useful Keyboard Shortcuts

Shortcut	What it does
`Tab`	Auto-complete command or filename
`Ctrl + C`	Cancel/stop current command
`Ctrl + D`	Logout or exit terminal
`Ctrl + L`	Clear screen (same as `clear`)
`Ctrl + R`	Search old commands
`/` Arrow	Go to previous/next command
`!!`	Run last command again

Important Interview Questions

Q1. Difference between `rm` and `rmdir`?

```
rmdir     Deletes EMPTY folders only
rm -rf    Deletes folders with all files inside (force delete)
```

Q2. What does `chmod 777` mean?

```
Gives Read + Write + Execute permission to:
  Owner
  Group
  Everyone

  Warning: This is a security risk!
```

Q3. How to run a program in background?

```
# Add & at the end of command
firefox &
sleep 100 &
```

Q4. How to see hidden files?

```
ls -a
# Hidden files start with . (dot)
# Example: .bashrc, .ssh, .config
```

Q5. What is full form of `sudo`?

```
sudo = Super User DO
It gives you admin/root power to run commands
```

One Page Cheat Sheet

```
# ===== Navigation =====
pwd                 # Where am I?
ls / ls -l / ls -a    # List files
cd folder           # Enter folder
cd ..               # Go back

# ===== File Operations =====
touch file.txt        # Create file
cp file1 file2        # Copy file
mv old new            # Move/Rename
rm file.txt           # Delete file
mkdir folder          # Create folder

# ===== View Files =====
cat file.txt          # Show content
head file.txt         # First 10 lines
tail file.txt         # Last 10 lines
```

```
less file.txt          # Scroll view

# ===== Permissions =====
chmod 755 file         # Change permission
chown user file        # Change owner
sudo command           # Admin access

# ===== System Info =====
df -h                  # Disk space
free -h                # RAM usage
top                    # Live monitor
uname -a               # OS info

# ===== Network =====
ip a                   # IP address
ping google.com        # Test network
ssh user@ip            # Remote connect
netstat -tulnp         # Open ports

# ===== Packages =====
sudo apt update
sudo apt install package_name

# ===== Process =====
ps aux                 # All processes
kill PID               # Stop process
systemctl start name   # Start service

# ===== Archive =====
tar -czvf file.tar.gz folder/    # Compress
tar -xzvf file.tar.gz            # Extract
zip -r file.zip folder/          # Zip
unzip file.zip                   # Unzip
```